

Homework 4

problem 2

part (a) $p(x) = \int p(x, z) dz \Rightarrow \log p(x) = \int \log p(x, z) dz$

part (b) $\log p(x) = \log \int p(x, z) \frac{q_\Phi(z|x)}{q_\Phi(z|x)} dz$

part (c) $\log p(x) = \log E_{q_\Phi(z|x)} \left[\frac{p(x, z)}{q_\Phi(z|x)} \right]$ $x \sim p(x), E[f(x)] = \int f(x) p(x) dx$

part (d) we could sample from $q_\Phi(z|x)$, so the expectation can be computed by Monte Carlo simulation. In that case, we don't need to compute $p(z|x)$ directly

part (e) $\log E[x] \geq E[\log(x)] \Rightarrow \log p(x) \geq E_{q_\Phi(z|x)} \left[\log \frac{p(x, z)}{q_\Phi(z|x)} \right]$ (lower bound)

part (f) we don't have to optimize $\log p(x)$, we just need to optimize ELBO, which is $E_{q_\Phi(z|x)} \left[\log \frac{p(x, z)}{q_\Phi(z|x)} \right]$

part (g) $E_{q_\Phi(z|x)} \left[\log \frac{p(x, z)}{q_\Phi(z|x)} \right] = E_{q_\Phi(z|x)} \left[\log \frac{p_\Phi(x|z) p(z)}{p_\Phi(z|x)} \right]$

$$= E_{q_\Phi(z|x)} (\log p_\Phi(x|z)) + E_{q_\Phi(z|x)} \left[\log \frac{p(z)}{p_\Phi(z|x)} \right]$$

$$= \underbrace{E_{q_\Phi(z|x)} [\log p_\Phi(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{KL}(q_\Phi(z|x) \| p(z))}_{\text{KL divergence}}$$

reconstruction term

KL divergence

$$D(p \| q) = E_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right] = E_{x \sim p} [\log p(x) - \log q(x)]$$

part(h) ① reconstruction loss: $E_{q(z|x)}[-\log p_{\hat{\theta}}(x|z)]$, it can let decoder learn how to generate data

② KL divergence: $E_{q(z|x)}\left[\log \frac{p(z)}{q_{\hat{\theta}}(z|x)}\right]$, it can let latent space similar to normal distribution, so we can sample from latent space

problem 3

$$\begin{aligned}\text{part (a)} \quad D_{KL}(P||Q) &= \mathbb{E}_P \left[\log \frac{P(x)}{Q(x)} \right] \\ &= - \mathbb{E}_P \left[\log \frac{Q(x)}{P(x)} \right]\end{aligned}$$

let $x = \frac{Q(x)}{P(x)}$, cause $\log(\cdot)$ is a concave function,
then $\mathbb{E}_P[\log x] \leq \log(\mathbb{E}_P(x))$

$$\therefore \mathbb{E}_P \left[\log \frac{Q(x)}{P(x)} \right] \leq \log(\mathbb{E}_P \left[\frac{Q(x)}{P(x)} \right])$$

$$\therefore \mathbb{E}_P \left(\frac{Q(x)}{P(x)} \right) = \int P(x) \frac{Q(x)}{P(x)} dx = \int Q(x) dx = 1$$

$$\therefore \mathbb{E}_P \left[\log \frac{Q(x)}{P(x)} \right] \leq \log 1 = 0 \Rightarrow \mathbb{E}_P \left[-\log \frac{Q(x)}{P(x)} \right] \geq 0 \Rightarrow D_{KL}(P||Q) \geq 0$$

part (b) $p \sim N(\mu_1, b_1^2)$, $q \sim N(\mu_2, b_2^2)$

In one dimension: $d=1$, $\Sigma_1 = b_1^2$, $\Sigma_2 = b_2^2$

$$\text{then: } \Sigma_2^{-1} = (b_2^2)^{-1} = b_2^{-2}$$

$$\text{tr}(\Sigma_2^{-1} \Sigma_1) = \text{tr}(b_2^{-2} b_1^2) = b_2^{-2} b_1^2$$

$$(\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) = (\mu_2 - \mu_1)^2 b_2^{-2}$$

$$\det(\Sigma_1) = b_1^2, \det(\Sigma_2) = b_2^2 \Rightarrow \ln \left(\frac{\det \Sigma_2}{\det \Sigma_1} \right) = \ln \frac{b_2^2}{b_1^2}$$

$$\therefore D_{KL}(P||Q) = \frac{1}{2} (b_2^{-2} b_1^2 + (\mu_2 - \mu_1)^2 b_2^{-2} - 1 + \ln \frac{b_2^2}{b_1^2})$$

part (c) step 1: $p(x) = \frac{1}{\sqrt{2\pi}b_1} \exp\left(-\frac{(x-\mu_1)^2}{2b_1^2}\right)$

$$Q(x) = \frac{1}{\sqrt{2\pi}b_2} \exp\left(-\frac{(x-\mu_2)^2}{2b_2^2}\right)$$

step 2: $\log \frac{p(x)}{Q(x)} = \log p(x) - \log Q(x)$

$$= -\frac{(x-\mu_1)^2}{2b_1^2} + \frac{(x-\mu_2)^2}{2b_2^2} + \log \sqrt{\frac{2\pi b_2^2}{2\pi b_1^2}}$$

$$= \log \frac{b_2}{b_1} + \frac{(x-\mu_2)^2}{2b_2^2} - \frac{(x-\mu_1)^2}{2b_1^2}$$

step 3: $D_{KL}(P||Q) = E_P\left[\log \frac{p(x)}{Q(x)}\right]$

$$= \log \frac{b_2}{b_1} + \frac{1}{2b_2^2} E_P[(x-\mu_2)^2] - \frac{1}{2b_1^2} E_P[(x-\mu_1)^2]$$

step 4: $E_P[(x-\mu_1)^2] = b_1^2$

($\because x \sim N(\mu_1, b_1^2) \Rightarrow \text{var}(x) = E[(x-E(x))^2]$)

$$E_P[(x-\mu_2)^2] = E_P[(x-\mu_1) + (\mu_1-\mu_2)]^2$$

$$= E_P[(x-\mu_1)^2 + 2(x-\mu_1)(\mu_1-\mu_2) + (\mu_1-\mu_2)^2]$$

$$= E_P[(x-\mu_1)^2] + 2(\mu_1-\mu_2) E_P[(x-\mu_1)] + (\mu_1-\mu_2)^2$$

$$= b_1^2 + (\mu_1-\mu_2)^2$$

conclusion: $D_{KL}(P||Q) = \log \frac{b_2}{b_1} + \frac{1}{2b_2^2} (b_1^2 + (\mu_1-\mu_2)^2) - \frac{1}{2b_1^2} \cdot b_1^2$

$$= \log \frac{b_2}{b_1} + \frac{1}{2b_2^2} (b_1^2 + (\mu_1-\mu_2)^2) - \frac{1}{2}$$

$$= \frac{1}{2} \left(\frac{b_1^2}{b_2^2} + \frac{(\mu_2-\mu_1)^2}{b_2^2} - 1 + \log \frac{b_2^2}{b_1^2} \right)$$